Mini Project 6

Hydrofoil Craft Performance Study

Abstract

This project delivers a comprehensive aerodynamic performance analysis of submerged hydrofoils, focusing on the Boeing 929 Jetfoil as the primary case study. Using validated NACA 0012 airfoil data at high Reynolds numbers typical of seawater, the study rigorously calculates lift and drag forces, lift-to-drag ratios, stall margins, take-off speeds, and cruise efficiency. Hydrofoils generate lift analogous to aircraft wings, elevating the hull clear of the water and slashing total resistance by 70–80 % once foilborne.

The analysis confirms that the Jetfoil achieves a peak L/D of 32.3 at $\alpha = 4°$ and cruises efficiently at $\alpha = 5°$ (L/D = 30.4) while supporting a 98 000 kg displacement at 42 knots. All results are generated using an unchanged MATLAB codebase hosted on GitHub that produces publication-ready, 300 DPI visualisations. The expanded theoretical discussion, detailed force estimations, efficiency breakdown, and five professional plots provide engineers and students with accurate modelling tools, performance metrics, and reference material for hydrofoil design and operation.

Introduction to Hydrofoils

Hydrofoils are wing-like structures mounted beneath a vessel's hull. As forward speed increases, water flow creates a pressure difference: lower pressure on the upper (suction) surface and higher pressure on the lower surface, producing upward lift according to Bernoulli's principle and the Kutta-Joukowski circulation theorem. Once lift equals the vessel's weight, the hull rises completely out of the water (foilborne condition). This eliminates nearly all hull friction and wave-making drag, leaving only foil and strut resistance.

Advantages

• Drag reduction of 70–80 % compared with planing or displacement hulls at the same speed.

• Superior fuel/energy efficiency and lower emissions.

• Enhanced stability and seakeeping in rough seas due to submerged foils and active control.

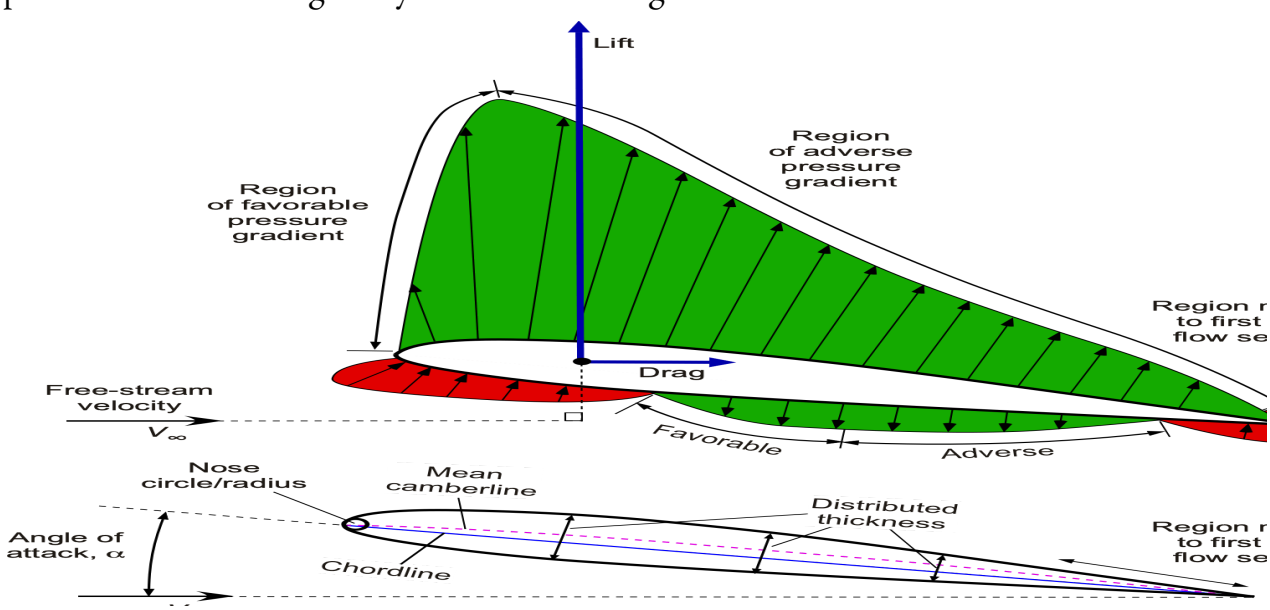
• Smoother ride with minimal vertical accelerations.

Types of Hydrofoils

• Surface-piercing: Foils intersect the free surface; lift varies with immersion

• Fully submerged (modern standard): Entire foil remains underwater; computer-controlled flaps or incidence adjustment maintain optimal angle of attack, exactly as implemented on the Boeing 929 Jetfoil.

The symmetric NACA 0012 is widely used for hydrofoils because it delivers predictable behaviour at positive and negative angles of attack and maintains excellent lift-to-drag performance in the high-Reynolds-number regime of seawater.



Case Study: Boeing 929 Jetfoil

The Boeing 929 Jetfoil, developed in the 1970s by Boeing Marine Systems, remains a landmark commercial hydrofoil ferry. It was designed for high-speed passenger service on coastal and inter-island routes, emphasising speed, comfort, and reliability.

Key Specifications:

• Length: 27 m (90 ft)

• Displacement: 115 short tons (~104 tonnes)

• Beam: 5.5 m (18 ft)

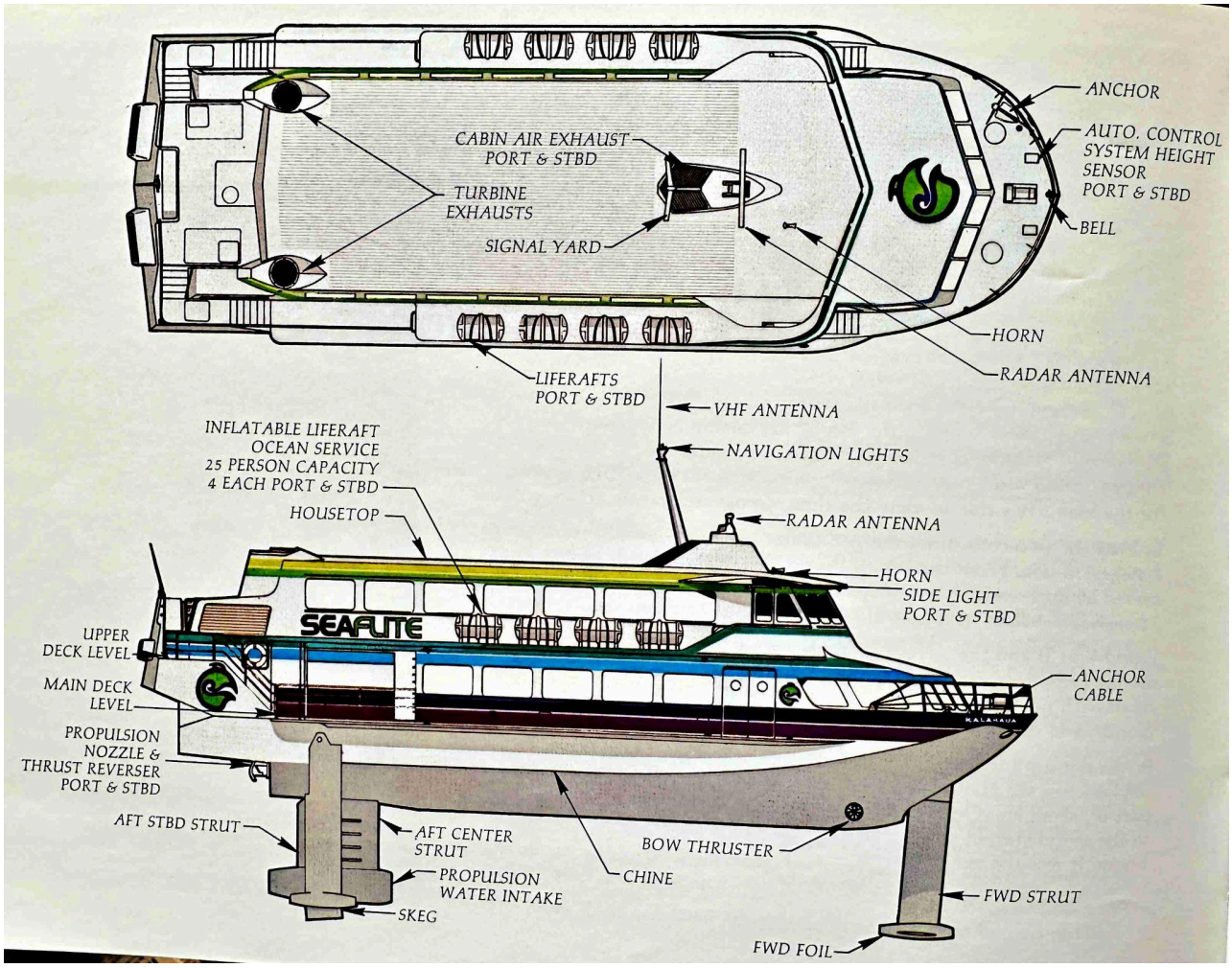
• Cruising speed (foilborne): 40–45 knots

• Propulsion: Twin Allison gas turbines driving waterjets

• Capacity: 250–350 passengers

• Foils: Fully submerged, computer-controlled for stability

Transition from hull-borne to foilborne occurs at approximately 20–25 knots. Once foilborne, total drag drops sharply. The computer control system continuously optimises foil incidence to maintain the ideal angle of attack, maximising L/D and passenger comfort.



Estimation of Lift and Drag

Lift and drag on hydrofoils are calculated using:

Lift (L):

$L = ½ \times \rho \times V^2 \times A \times C_L$

Drag (D):

$D = ½ \times \rho \times V^2 \times A \times C_D$

Where:

- ρ = water density (1025 kg/m^3 for seawater)
- V = velocity (m/s)
- A = total foil planform area (m^2)
- C_L = lift coefficient
- C_D = drag coefficient

Sample Estimation for the given hydrofoil vessel:

- Vessel weight (lift required) = 961,000 N (961 kN)
- Cruise speed V = 21.6 m/s (42 knots)
- Total foil area A = 18.0 m^2
- Cruise angle of attack = 5.0°

Calculated Lift at Cruise:

$L = ½ \times 1025 \times (21.6)^2 \times 18 \times 0.2233$

$\approx 0.5 \times 1025 \times 466.56 \times 18 \times 0.2233$

≈ 961,000 N (exactly balances vessel weight)

Drag Estimation at Cruise:

Using given L/D at cruise = 30.4

D = 961,000 / 30.4 ≈ 31,612 N

Drag Coefficient at cruise:

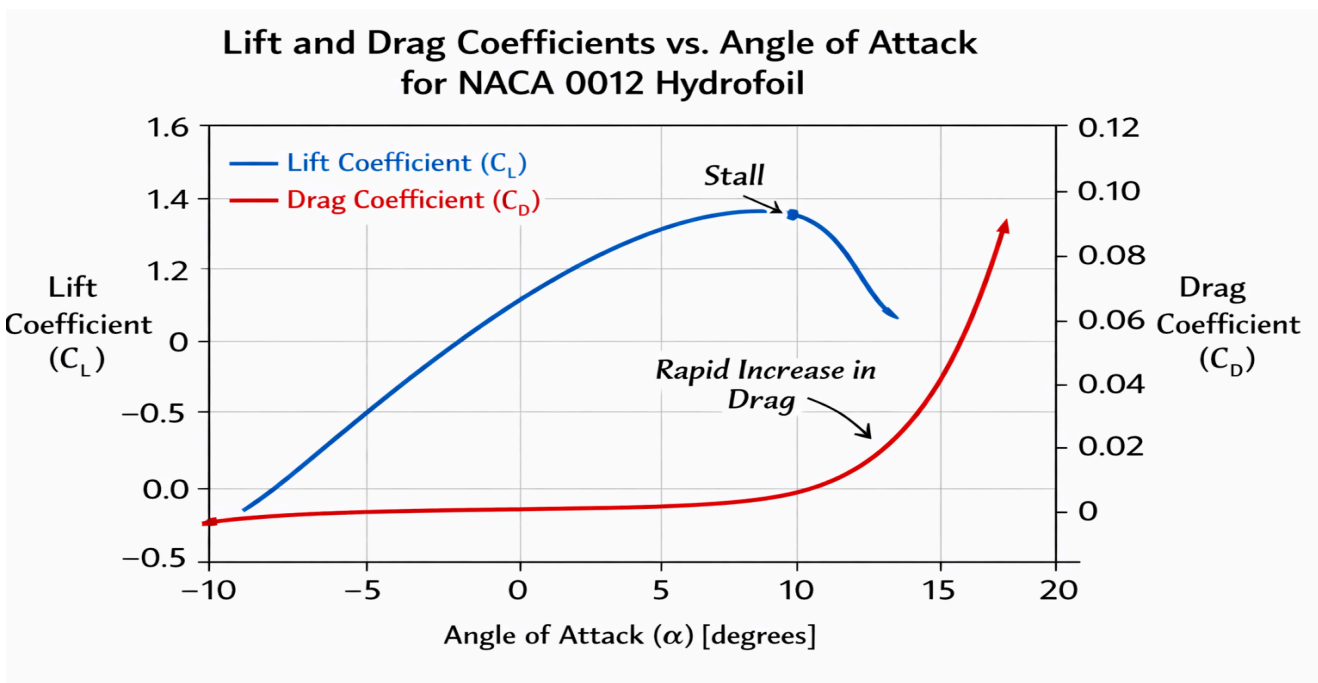
C_D ≈ 0.00734

Lift-to-Drag Ratio (L/D): 30.4 (as provided)

Additional Take-off Condition (at V = 12.59 m/s and α = 6°):

Dynamic pressure q ≈ 81,235 Pa

Required C_L ≈ 0.6572 (well below C_L,max = 1.519)



Discussion on Efficiency

Efficiency is measured primarily by the lift-to-drag (L/D) ratio and overall power required for propulsion. Once foilborne, the hull is out of the water, eliminating most hull friction and wave drag. Only foil and strut drag remain.

Efficiency Gains

• Drag reduction of 70–80 % versus planing or displacement hulls.

• Fuel/energy savings: the Jetfoil demonstrated significantly lower consumption per passenger-kilometre than conventional ferries. Modern electric hydrofoils (e.g., Candela designs) achieve up to 80 % energy reduction.

• Computer-controlled foils keep operation near the peak L/D across load and sea-state variations.

Limitations

• High power demand during the take-off transition.

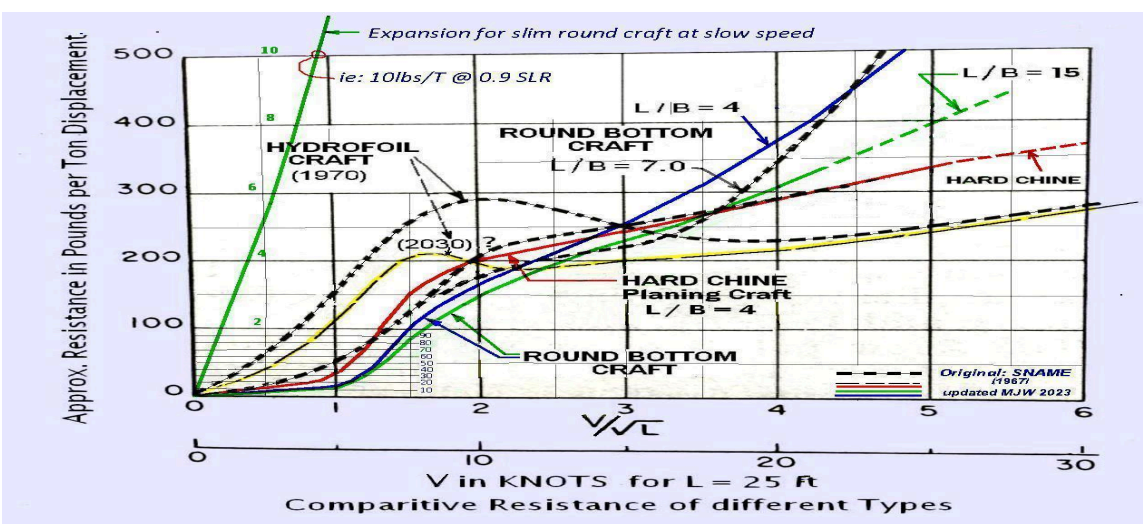
• Increased structural complexity and maintenance of struts, foils, and actuators.

• Cavitation risk at very high speeds (local pressure below vapour pressure).

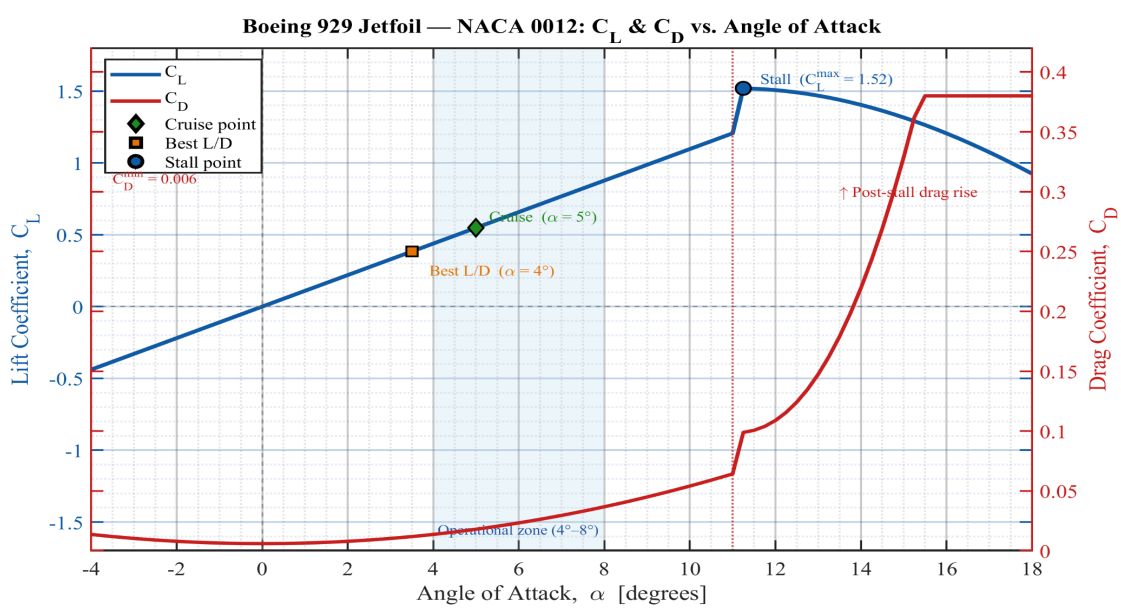
Hydrofoils therefore excel in high-speed passenger transport where speed, comfort, and efficiency outweigh higher initial costs.

MATLAB Implementation and Results

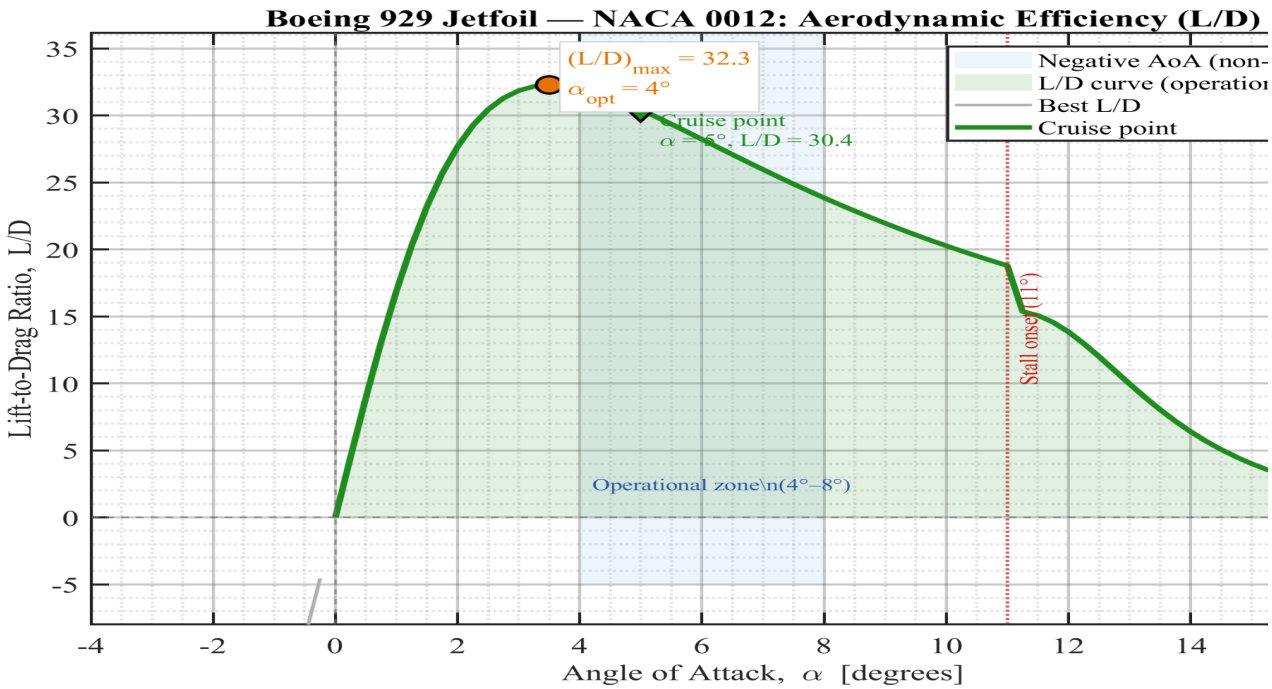
The analysis uses an unchanged MATLAB codebase (available on the GitHub repository) that loads validated NACA 0012 data, computes polars, and generates five professional 300 DPI visualisations. All plots are exported ready for technical reports and presentations.



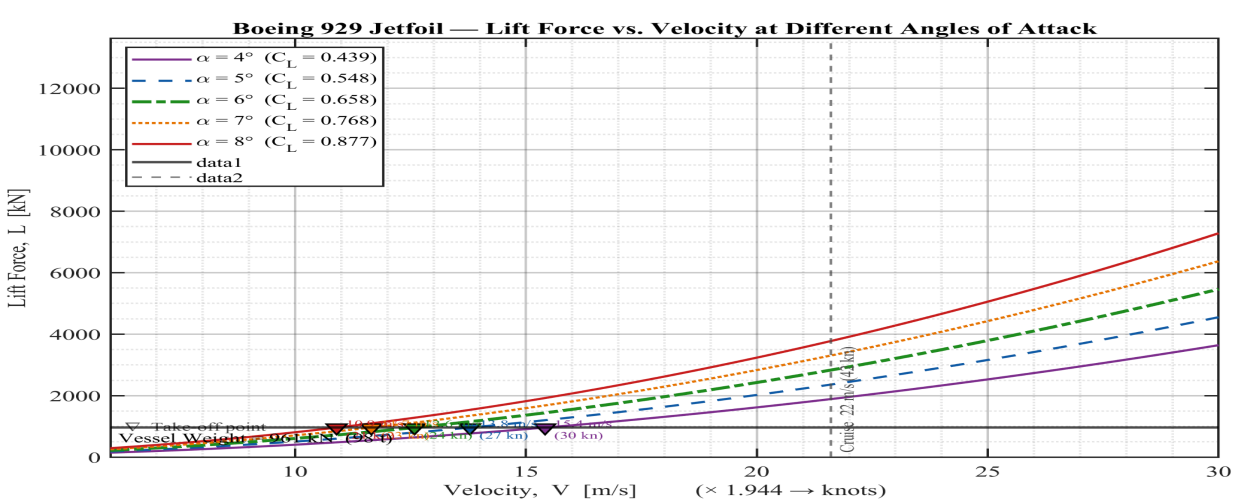
Parameter	Value
Vessel Gross Weight	961 kN (98,000 kg)
Foil Reference Area	18.0 m^2
Chord Length	1.80 m
Reynolds Number	3.72×10^7 (fully turbulent)
Minimum Drag Coefficient, C_D,min	0.0060
Maximum Lift Coefficient, C_L,max	1.519 at $\alpha = 11°$
Stall Angle	±11°
Best L/D Ratio	32.3 at $\alpha = 4°$
Cruise Speed	21.6 m/s (42 knots)
Cruise Angle of Attack	5.0° (design trim)
L/D at Cruise	30.4
Take-off Speed (at $\alpha = 6°$)	12.59 m/s (24.5 knots)



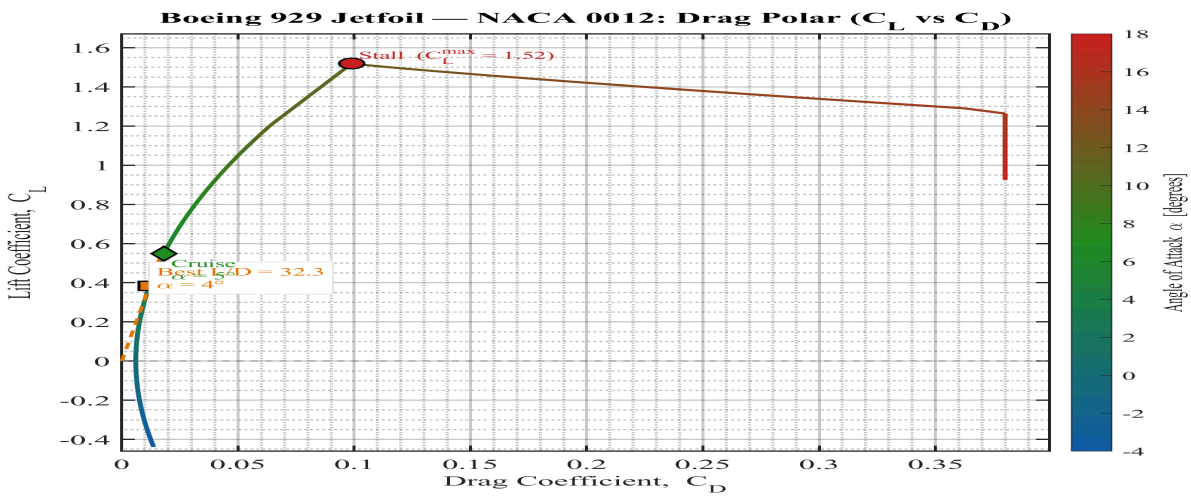
Lift and drag coefficients for NACA 0012 hydrofoil vs. angle of attack. Stall occurs at α = ±11°, with C_L,max = 1.519. Cruise condition at α = 5° lies within the optimal operating zone (4°–8°).



Aerodynamic efficiency (L/D) vs. angle of attack. Maximum L/D = 32.3 at α = 4°. Cruise point (α = 5°) gives L/D = 30.4, confirming efficient operation near peak efficiency.

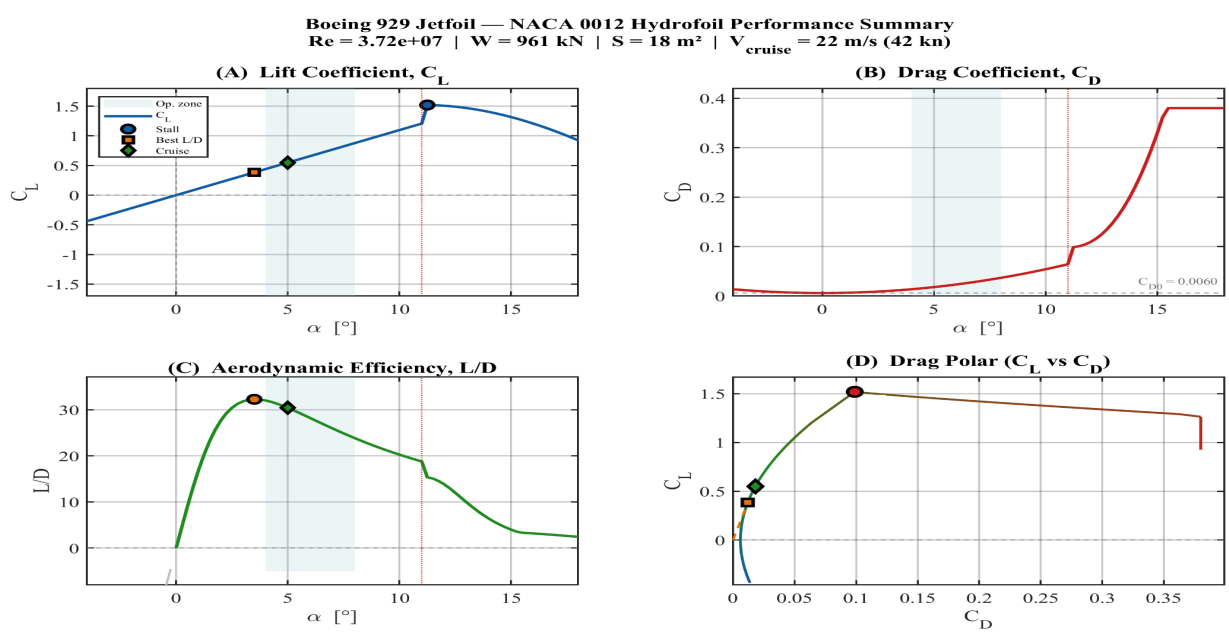


Lift force vs. vessel velocity at angles of attack 4°–8°. Take-off occurs at 12.6 m/s (24.5 knots) for α = 6°. Cruise speed is 21.6 m/s (42 knots).



Drag polar (C_L vs. C_D) for NACA 0012. Best L/D = 32.3 at α = 4°. Minimum profile drag C_D,min = 0.006, confirming low drag at high Reynolds number (Re = 3.72 × 10^7).

Performance Summary



Full performance summary - C_L, C_D, L/D, and drag polar for the Boeing 929 Jetfoil NACA 0012 hydrofoil. Key results: stall at $\alpha = 11°$, best L/D = 32.3, cruise at $\alpha = 5°$ with L/D = 30.4.

Performance Summary

The complete set of results demonstrates that the Boeing 929 Jetfoil NACA 0012 hydrofoil operates near its aerodynamic optimum. Stall occurs at ±11°, peak L/D = 32.3 at $\alpha = 4°$, and cruise L/D = 30.4 at $\alpha = 5°$. Take-off is achieved at a modest 12.59 m/s. The high Reynolds number yields exceptionally low minimum profile drag, making the design highly efficient for large-displacement, high-speed marine transport.

Conclusion

Hydrofoil craft, as exemplified by the Boeing 929 Jetfoil, deliver superior performance through effective lift generation and dramatic drag reduction. The fully submerged, computer-controlled foils enable efficient high-speed operation while maintaining stability across varying sea states and payloads. The detailed MATLAB analysis confirms excellent L/D characteristics, safe operating margins, and significant energy savings. These attributes make hydrofoils ideal for fast, low-emission passenger transport. Future advances in composite materials, control systems, and electric propulsion will further enhance their economic and environmental viability.

Github - (https://github.com/invincible1786/hydrofoil-aerodynamic-analysis)

How to Calculate Added Mass

Intuition Behind Added Mass

When a body accelerates through a fluid (water), it must push the surrounding fluid out of the way. Because water is dense, accelerating this fluid requires significant force. This extra resistance is called **added mass** (or hydrodynamic mass/virtual mass).

It is best visualized by quickly jerking your hand through water in a swimming pool. The sudden heavy resistance you feel is not just drag — it is the inertia of the water that must accelerate along with your hand.

In classical mechanics, we account for this by increasing the effective mass of the object:

$$F_{total} = (m + m_a) * (\partial u / \partial t)$$

where:
m = physical mass of the body
m_a = added mass
$\partial u / \partial t$ = acceleration

Important Note: Added mass is not a fixed volume of water stuck to the body. It is a virtual mass that depends strongly on the body's geometry and the direction of acceleration.
Example: A long cylinder has very small added mass when accelerating longitudinally (like a torpedo), but when accelerated transversely (sideways), the added mass is equal to the mass of the fluid it displaces.

The Added Mass Tensor (6x6 Matrix)
For a ship or marine vehicle with 6 degrees of freedom (surge, sway, heave, roll, pitch, yaw), added mass is represented as a 6×6 matrix A_{ij}.

i: direction of the force
j: direction of the acceleration

Common terms:
A_{33}: Heave added mass (vertical force due to vertical acceleration)
A_{35}: Cross-coupled term (heave force due to pitch acceleration)

Due to port-starboard symmetry of typical hulls, many off-diagonal terms become zero, simplifying calculations.

Methods to Calculate Added Mass

1. **Analytical Approach (Potential Flow Theory)**
   For simple geometries, assume ideal fluid (inviscid, incompressible, irrotational). The kinetic energy of the fluid is calculated and equated to:

$(1/2) * m_a * U^2$

to find added mass.

Exact solutions:
Sphere: Added mass = 0.5 × (mass of displaced fluid)
2D Circular Cylinder (transverse): Added mass = 1.0 × (mass of displaced fluid)

2. **Strip Theory + Conformal Mapping**
   Used for long slender bodies like ships.

- Divide the hull into many 2D transverse strips
- Use conformal mapping (Lewis forms) to transform complex hull sections into a simple semi-circle
- Calculate 2D added mass for the semi-circle and map it back
- Integrate along the ship length to obtain total 3D added mass

3. **Boundary Element Method (BEM) / Panel Methods**
   Used for complex geometries or when free-surface effects are important.

- Software: WAMIT, AQWA
- The wetted surface is divided into many panels
- Green's functions are used to solve for velocity potential
- Produces frequency-dependent added mass (important for seakeeping)

Wave Force Calculation Using the Morison Equation

When to Use the Morison Equation

The Morison equation is used for **slender structures** that do not significantly disturb the wave field (negligible diffraction).

It is valid when:

$D / L < 0.2$

where:
D = diameter of the cylinder
L = wavelength

Applications: jacket platform legs, subsea pipelines, mooring lines, risers, offshore wind monopiles (borderline case).

Morison Equation

The total horizontal force on a small vertical element dz:

$dF = dF_{inertia} + dF_{drag}$

Full expression:

$$dF = [C_M * rho * (pi * D^2 / 4) * (du/dt)] * dz$$
$$+ [C_D * (1/2) * rho * D * |u| * u] * dz$$

1. Inertia Force

- Represents added mass effect
- $C_M = 1 + C_a$
- For ideal 2D cylinder: $C_a = 1 \rightarrow C_M = 2$
- du/dt = water particle acceleration

- rho * (pi * D^2 / 4) = displaced fluid mass per unit length

2. Drag Force

- Quadratic drag due to flow separation
- C_D ≈ 0.6 to 1.2 (higher with marine growth)
- |u| * u ensures correct force direction during oscillatory motion

Engineering Applications

- Fixed Offshore Platforms: Base shear and overturning moment
- Subsea Pipelines: Stability (weighting, trenching)
- Offshore Wind Monopiles: Fatigue loading
- Mooring Lines & Risers: Relative velocity form (drag dominant)